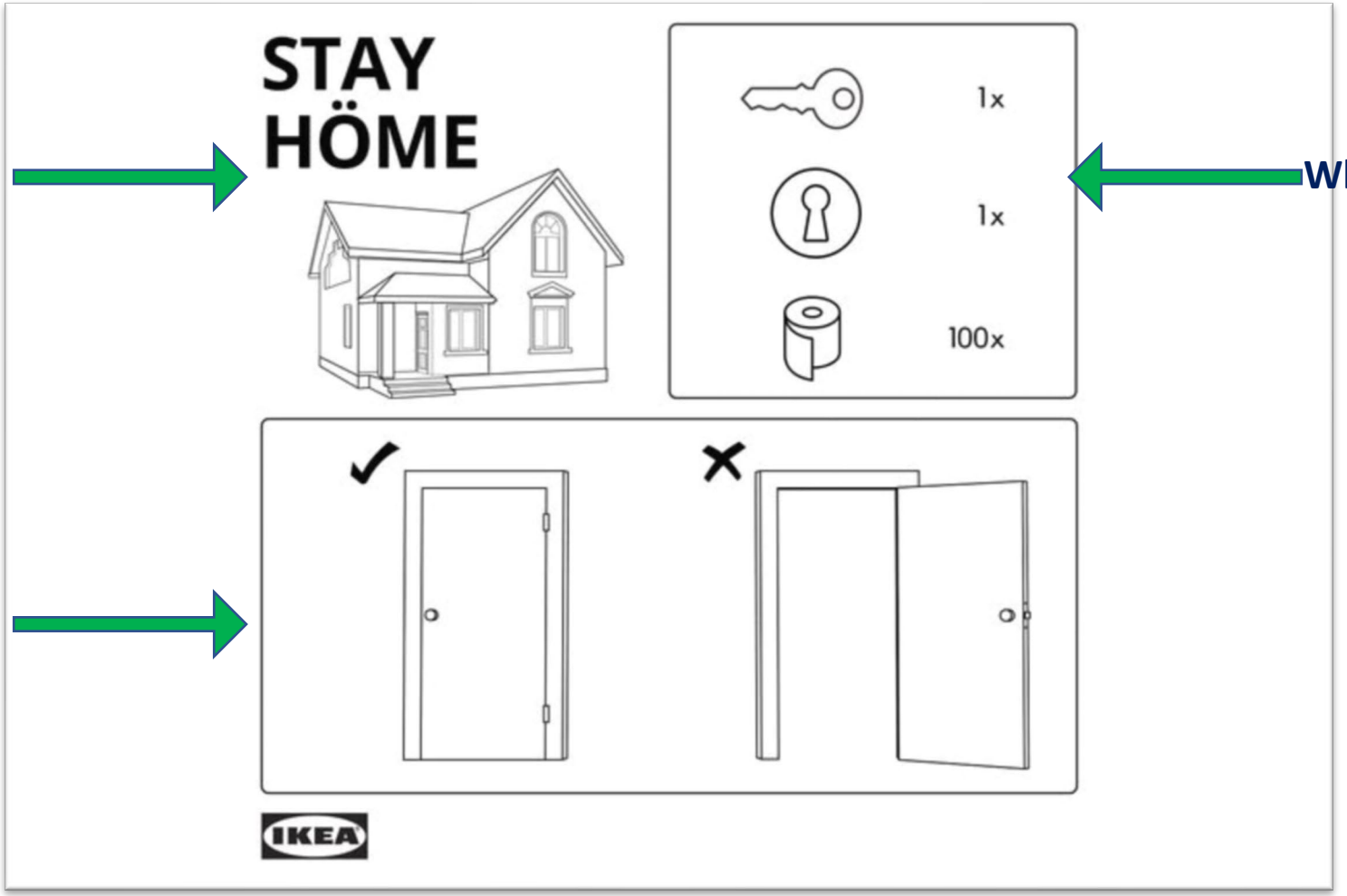
Writing user documentation

Some instructions are easy to follow

Ultimate Goal

What user needs to achieve goal

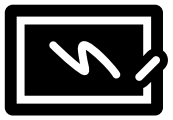
Successful outcome



[1] Ikea Image: Stay Home

Trying instructions out can help us.....

Please follow the instructions alone.



You will need a pen and paper or tablet (something to draw on and show)

Give yourself 5-10 minutes to complete the instructions.

Don’t share. I’ll let you know what to do when time is up.

Share your creations when asked

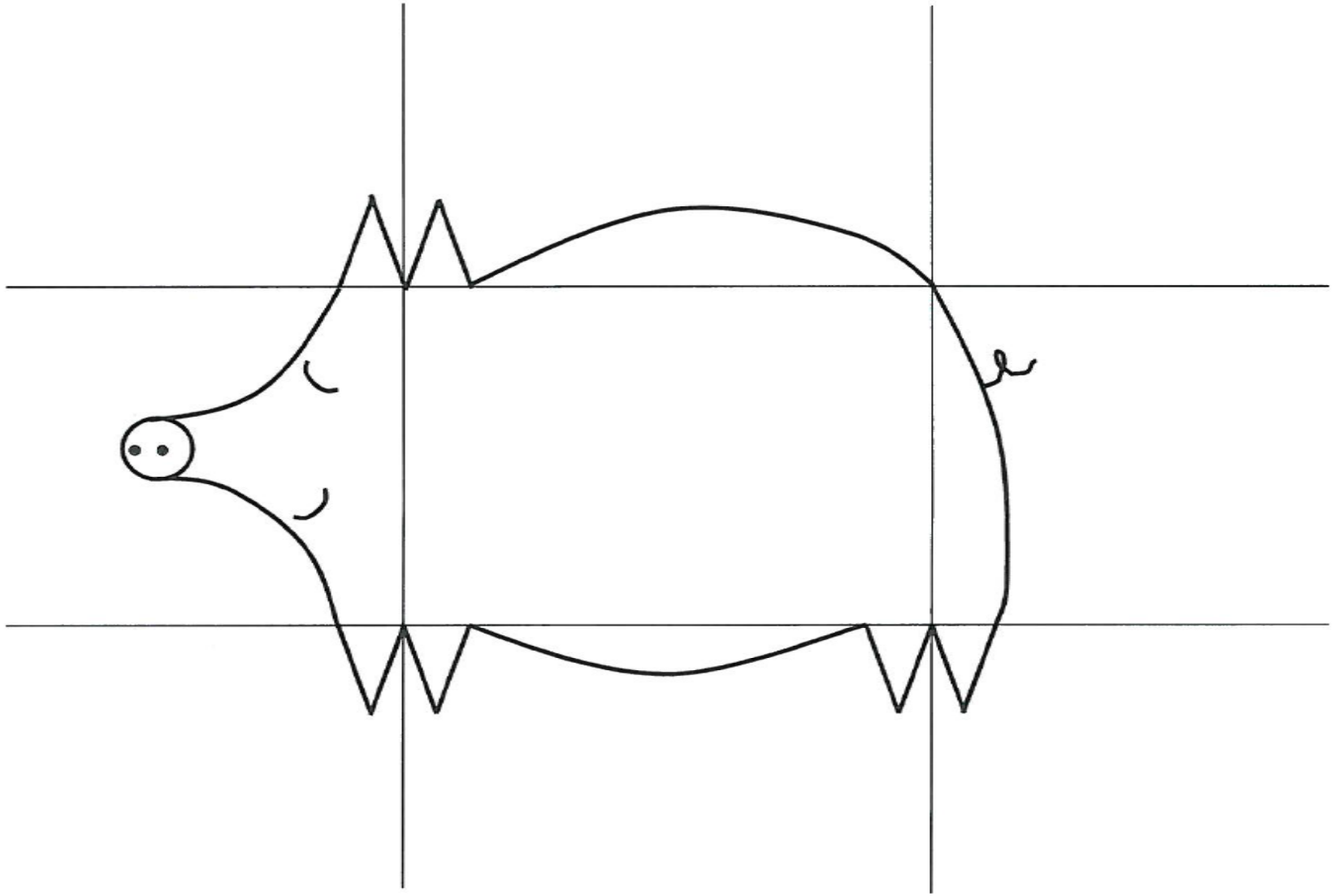
Some differences may occur.

These differences are expected.

I’ll show you what you should have created on the next slide.

Time to share.

This is how it should look: like a pig!



Problems with these instructions

- ? Don’t know what we are meant to create
- ? No numbers
- ? Wording poor/ Terms are too vague (curly-cue!)
- ? No standard for writing instruction – verbs

Your future may include writing user docs

Instructions for future developers, users, curious readers

Entire in-house manuals, specific job tasks

All instructions must be effective and helpful for users

As ever, focus on the reader

- Reader might use the instructions *once or several times*
- Instructions must be *useful* for first-time readers and repeat readers
- The instructions must be *at the level* of the intended reader

Your reader is a student entering CST in September 2026.

Writer vs. user experience level

- Instructions are written by technically proficient people for non-technically proficient people
- Writers overestimate reader knowledge and abilities
- What’s obvious to the writer isn’t obvious to the reader

Clear only if known issue

Writer reviews procedures; instructions make perfect sense to them.

Instructions are clear because they are *known* to writer.

Instructions may not be clear to reader as the info is *unknown*

Instructions must be tested to ensure effectiveness

Cures for lack of clarity

- Give instructions to a test audience and see if clear to them
- Read the instructions aloud
- Write early and allow time between writing and editing

Understanding your users: a key step

- State level of *user* that you are writing for
- State level of *pre-knowledge* to complete your instructions

Task analysis for user

Imagine you are designing a new VSCode.

List user tasks for each user type:

Simple, frequent tasks for novice users (your audience)

One off tasks for casual users

Complex, infrequent tasks for advanced users

Summary

- Think of and create user level of knowledge and needs
- Test instructions with partner and peers
- Examine knowledge difference between you and the user

Test sample as we look at assignment

Go through sample with your partner.

Write down any key information you want to apply to your future docs

1. Use an overall introduction: test yours

- State purpose of software and purpose of specific instructions
- State who instructions are aimed at (users and user level/ability)
- State equipment user needs to complete the instructions

  (Windows version? Something already installed?)
- Add notes, cautions and warnings that apply to *whole instructions*
- Add typographical conventions you will use to guide readers

What did you notice with sample intros?

Some good?
Some not so good?

What your group needs to remember. Note it down.

Always start with the intro as this sets you up as writers.
It provides your style guide.
Start tomorrow in lab

Heading identifies purpose.

How to Change the Size or Resolution of Internet Graphics for Print Resolution

The following instructions show you how to change an Internet graphic from screen resolution to print resolution.

Internet graphics are optimized for quick download by having their resolution reduced to 72 dpi (dots per inch), which is the resolution at which monitors display information. However, when you download Internet graphics and print them (for instance in a report) the 72 dpi resolution will look very pixilated. You, therefore, need to change the resolution of Internet graphics to 150 dpi before inserting them into documents you intend to print on paper.

These instructions assume that you will be using Adobe Photoshop CS, the digital editing software available in your computer labs in NE1. Other digital editing software packages offer the same functions, but call them by different names and sort them differently in their menu systems.

Introduction:

- clarifies purpose
- explains necessity of following instructions
- states any assumptions

Conventions

These instructions use the following typographic conventions:

Dialog Window Buttons and Keyboard Commands:	**Copy, Paste, Save As, Okay**
Menu selection sequences:	[Start] > [Programs] > [Adobe] > [Adobe Photoshop CS]
Dialog Windows	"New File Dialog Window"

Intro states conventions, so the writer can refer to selection sequences and keyboard and mouse commands without creating confusion.

Conventions allow the writer/reader to identify commands like *Copy* and components like dialogue boxes instantly.

2. The actual How-To: creating steps

A step is something for the reader to do

Steps are actions only; they must be separated from information

✕ information *does not* give reader anything to do

Steps are numbered

One user action = one step

3. Separate steps (actions) from information

A step is something for the reader *to do*. Steps are actions only. Steps are numbered.

If there is no action, it is another feature of the instructions. Therefore, it should not be numbered.

Step = action

No action.
Line space indicates separation from Step. This is often **called a feedback statement**

3. Open a new Photoshop file: [File] > [New].

At this point, the Photoshop "New File Dialog Window" will open. Please note that Photoshop will automatically create a file that matches the size and resolution of the graphic in the Windows Pasteboard.

Very poor step: actions underlined

1. Install Windows using Boot Camp. Insert Windows disk and reboot. Hold C when you hear the beeping sound. Windows installation should start, follow the usual installation process of Windows. After Windows is installed, insert your OSX disk and it should install all your Apple drivers and files. Update your Apple Software (there will be an option on the start menu to do so).

4. Number all steps

Only steps are numbered

Other parts gives reasons, for example, as well as necessary information that support the steps

Instruction steps are not bolded because bolding identifies software commands.

Notes in a separate paragraph.

Note the consistent use of “At this point” to introduce the *results* of an action.
Easier for users to scan and comprehend the instructions.
Always look for ways to make your writing more consistent.

3. Open a new Photoshop file: [File] > [New].

At this point, the Photoshop “New File Dialog Window” will open. Please note that Photoshop will automatically create a file that matches the size and resolution of the graphic in the Windows Pasteboard.

4. Change the default file name to a file name that will make it easy for you to identify the contents of the file.

5. Click Okay.

At this point, you will see a blank file. Photoshop is waiting for you to input content.

6. Paste the contents of your Windows Pasteboard into the Photoshop file: Ctrl + V (paste command).

At this point, you should see your image in the file window. You now need to change the resolution of this image.

5. Start steps with imperative voice

Also known as command voice

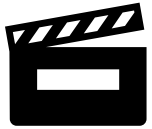
Verbs are commands:

- Click, or even Left-Click
- Press
- Enter

Not: “You should click on…”

Command: “Click on the tab and…”

How well does your sample follow these specs?



- Steps are actions only
- Check they start with command verbs
- Check they are separated from information/feedback statements.

5. Use visuals effectively

Include graphics to show

- the *actions* you want the user to perform
- *the specific* area of the screen in which the user should be looking
- the *interface or data* the user should be manipulating

Graphics help the users understand if they are following the instructions correctly

Graphics need alt-text and labels for accessibility (markdown)

Place as close to step as possible: see next slide

Changing the resolution of a graphic

When you find an Internet graphic you want to use in your report, please do the following:

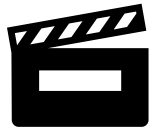
1. Right click on the graphic and select *Copy* from the "Windows Pop-up Menu."

2. Open Adobe Photoshop CS: [Start] > [Programs] > [Adobe] > [Adobe Photoshop CS].

   Adobe Photoshop is a memory intensive program and will take some time to load. Once in Photoshop,

How well do your samples use visual info?

In some cases, the visual might be code.



Is the visual in the right place?
Is a visual needed?
Is the visual helpful?

Share what you notice about use of visuals here with partner.
Make notes about using visuals

6. Use notes, cautions (admonitions)

Notes, cautions, and warnings prevent users from making mistakes

Place directly before the instruction step to which it applies

Place at beginning of section if it applies to the whole section

Seriousness of the warning should be reflected in the formatting

Should use bold for the title (**Note: Caution: Warning**:) and regular text for information

Notes, cautions, warnings

Notes, cautions, warnings, and danger tags should be formatted with increasing emphasis.

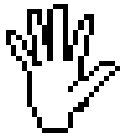
Readers should be in no doubt as to the threat level.

Depending on the hazards the reader is likely to encounter, only three or even only two levels of notes may be necessary.

Notes replace bold and underline

Show important information by visually distinguishing it from the rest of the text.

Often preceded by icons such as a raised hand. The word Note is commonly bolded.



Note: In all types of manuals, notes point out something of special interest or importance to the reader. Failure to read the note will not result in physical harm to the reader, equipment, or data.

Use tabs/indents to make visually distinguishable from text
Use emojis/icons in GitHub for your icons

Cautions draw more attention

Bring attention to anything that could damage equipment or cause data loss.

Show what could happen if the caution is ignored.

Always place the caution before the step it applies to.

Very common to use an ‘attention’ icon beside a caution.

Caution: Do not turn the monitor off by unplugging it from the computer or wall socket. Severe damage to the monitor may result. Turn the monitor off before unplugging it.

Warnings are more severe cautions

Draw special attention to anything that could injure or kill the reader.

Always placed before the step in the procedure they relate to

Use attention icon.

WARNING: Unplug the computer before removing the outer case. If you don't, severe electrical shock may result.

Notice the capitals.

Notice the indentation of text

How well does your sample use admonitions?

They need to appear on the Intro: do they?

How well are they placed? They should be before user does something.

How are they helping you as the reader/user?

7. Introduction for every task page

Intro (also known as overview) follows a heading

The intro/overview lets user know what they can expect to achieve on this page.

It adds any admonitions that relate to the whole page

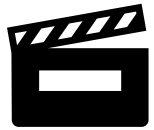
8. Conclusions for every task page

readers unsure if they've performed procedure correctly.

Provide a quick check to reassure them:

"If you have followed the instructions, the status light and all four hard drive lights on the NAS should be steady green."

How well does the sample use intros and conclusions?



Are the intros clear and helpful?

Is anything missing that could be added?

How do conclusions let you know you have been successful?

Do conclusions show you what success is? Do they need to?

What do you notice about its use of intros and conclusions?

Add any notes for your group to help you with your docs.

9. Use a glossary to define terms

Alphabetical listing of terms and definitions specifically from your instructions

Terms not familiar to the users but essential to understanding instructions and concepts

Accurately identify the users and write meaningful definitions

Python Glossary

- *Portability*: a property of a program that can run on more than one computer.
- *Print statement*: an instruction that causes the Python interpreter to display a value on the screen.
- *Prompt*: characters displayed by the interpreter to indicate it is ready to take input from the user.

How well does the Glossary function?

Are the terms helpful?
Are they clear?
Are they connected to the actual instruction content?
What else do you notice?

Make notes for your group about Glossary.

Sometimes users make mistakes

Add a troubleshooting guide later in the document:

“If none of the lights are on, check that …”

8. Troubleshooting Guide

Solutions for common problems in your instructions

Lists appropriate solution/action to the problem

Not necessarily comprehensive, nor mandatory, but saves time and tech support costs

Ubuntu Installation Troubleshooting Guide

- If the kernel hangs during the boot process, check the boot parameters.

If drivers are not recognized...

If you complete the boot phase but cannot complete the install...

How helpful is the troubleshooting?

Are the situations clear?
Are the solutions clear?
What else do you notice?
Are they directly related to the instructions?

Make notes for your group.

You’ve checked key elements

No matter how clear instructions are

THEY MUST BE TESTED TO SEE IF YOU CAN COMPLETE THEM

Now, try to complete the instructions

One person try, speak out as you try

One person make notes of what worked and what didn’t

- Was all information available/clear?
- Was info missing?
- Did you need more images?
- Did you need anything else?
- What helped you to complete the instructions?
- What did the instructions need to do better to help you?

Your assignment.

You have partners. If you don’t, you soon will!

Choose a piece of software (not an app) and look at some functions of the software to create instructions for. You are making a “How to do x using y” guide

User should need instructions to be able to complete task.

Remember, user is a newbie in CST. Remember September 2025. What would you liked to have instructions for?

Add software name to GDocs to confirm your project.

Key to assignment: human in the loop

- We can use tools like AI to help write things (next section in lecture)
- The human in the loop needs to verify the output of the work
- Always add the human in the loop at points in your process
- You decide where this needs to be
- I'd place it in as many points as possible so you stay on track.

Questions?

No matter how small, ask away.

Use time in class

DM on Discord

Walkthroughs

Try the software task out before you write any instructions

Enables you as writer to understand

- graphics needed (and which part of graphic needs focus)
- steps needed
- notes and cautions needed

Enable you to see all elements of walking through tasks so you don’t miss a key part that could affect user performing the task

Use a Google document to help

Tasks/Steps/Actions	Graphics	Notes

Go through each of the steps of a task and type information in the Google Doc (on its way)

Work out which images are needed for the steps.

Write info; *don't add the image* but write what the image should be.

Add information in notes area, such as if notes are needed, if screenshot should be cropped, for example.

Next steps: homework!

1. Read assignment
2. Choose software
3. Choose and plan specific software tasks
4. Go to MKDocs Material and follow installation video carefully
5. Add partner and me to your repository

References

[1] McCann Advertising Agency: Ikea Stay Home. Retrieved from https://www.adsoftheworld.com/campaigns/stay-home-cd646fd6-f986-4ebf-9ddf-6611a40ddf8d

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